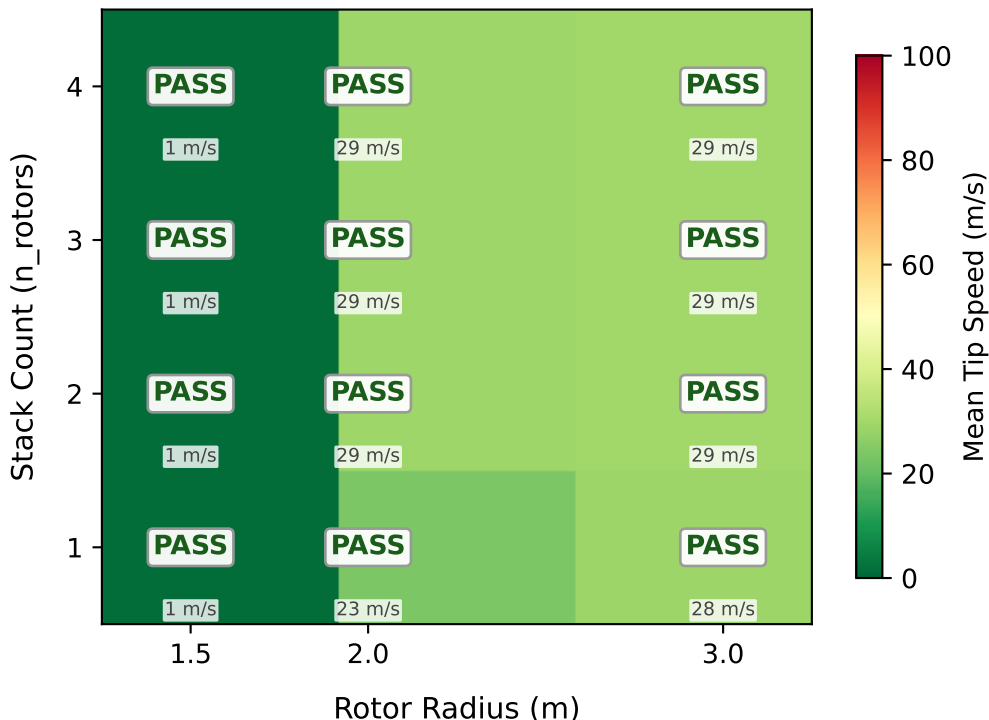


Tip Speed — Noise Gate (≤ 100 m/s)



IMPLICATIONS: All configurations stay well below 100 m/s tip speed — the noise gate does not constrain this design space. Tip speed saturates at ~ 30 m/s for $R \geq 2.0$ m: larger rotors spin slower, self-regulating to the same tip speed via BEM equilibrium. $R = 1.5$ m is stalled — $Re = 15k$, below the transitional threshold. Blades bottom out at 10 RPM regardless of stack count. Noise is not a limiting factor at these scales. Data: BEM v2.0, corrected polygon solver. See SPEC.md §6.6.